
Swiss Knife

Decode-Time Alignment via Tournament Sampling

A critical analysis: positioning, novelty, implementation

Lab	Pragya Lab, BITS Pilani Goa
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Contents

1	Executive Summary	2
2	The Decode-Time Alignment Landscape	3
2.1	What exists and what each method does	3
2.2	The three closest rivals, in detail	3
3	What Swiss Knife Actually Contributes	5
4	Tournament Sampling: The Mechanics	6
4.1	Why pairwise comparisons?	6
4.2	Format 1: Knockout bracket	6
4.3	Format 2: Swiss-system schedule	7
4.3.1	How Swiss-system works	7
4.3.2	Pairing rule in each round	7
4.3.3	Complexity	7
4.3.4	Why use Swiss-system over knockout for Swiss Knife	7
4.4	The match score function	8
5	Option A: Non-Speculative (Best-of-K Tournament)	9
5.1	What Option A is honest about	9
5.2	Why Option A is strictly weaker than Option B	10
6	Option B: Speculative-Decoding-Integrated Tournament	11
6.1	Recap: standard speculative decoding	11
6.2	The reframing: verifier as alignment socket	11
6.3	What makes this actually speculative	11
6.4	Compute profile	12
6.5	Architecture diagram	13
7	Fitting Swiss Knife into the Alignment Framework	14
8	Implementation Plan	15
8.1	The one-sentence novelty claim	15
8.2	Revised priority order	15
8.3	Decision points	15
8.4	The question reviewers will ask, with your answer	16
9	Summary Table: What to Do Next Week	17

1 Executive Summary

This document re-examines the Swiss Knife proposal against the current decode-time alignment literature (as of early 2026) and identifies three issues with the draft framing, along with the corrective path.

1. **The multi-head framing is not novel in itself.** MOD (Shi et al., NeurIPS 2024), GenARM (Xu et al., ICLR 2025), PARM (2025), Collab (2025), and several others already operate in the “frozen generator + swappable objective modules” regime. The claim “decode-time alignment with modular heads that can hot-swap” is taken.
2. **The genuine novelty is the *tournament selection rule*.** Existing methods combine objective scores either by linear mixture (MOD) or by argmax (ARGS, DeAL). Swiss Knife proposes pairwise tournament aggregation, which is mechanically different and provides a specific defense against reward-maximisation degeneracies (refuse-always, boilerplate collapse).
3. **The “speculative decoding” framing in the abstract is currently aspirational.** The algorithm described in the plan document is Best-of-N with a tournament selector; it is not speculative decoding in the sense of Leviathan et al. (2023). Two honest paths exist: (A) drop the speculative framing and position as decode-time tournament alignment, or (B) actually build the speculative-decoding-integrated version where the auditor replaces the verifier. This document recommends path B, and details it.

Key insight. The paper that works is: “Tournament-based decode-time alignment, implemented as a replacement verifier step inside a speculative-decoding loop, with a pluggable objective blade.” Every word in that sentence is doing load-bearing work, and the experiments must defend each.

2 The Decode-Time Alignment Landscape

2.1 What exists and what each method does

Table 1 catalogues seventeen relevant methods along the dimensions that actually matter for positioning Swiss Knife. The axes are: (a) what the method combines, (b) how it combines them, (c) whether it supports multiple objectives, (d) whether objectives can be hot-swapped post-deployment, (e) whether it has any defense against reward-max degeneracies, and (f) whether the generator backbone is frozen.

#	Method	Year	Combination rule	Multi-obj?	Hot-swap?	Frozen?
1	ARGS	2024	Draft logit + $\lambda \cdot r$, argmax	No	No	Yes
2	DeAL	2024	Beam search + reward rerank	Partial	Prompt-only	Yes
3	CARDS	2024	Span-level rejection sampling	No	No	Yes
4	Transfer Q*	2024	Learned Q-function guidance	No	No	Yes
5	MOD	2024	Linear mixture of next-token distributions	Yes (weight w)	Yes	Yes
6	GenARM	2025	Token-level ARM reward addition	Yes (multi-ARM sum)	Yes	Yes
7	PARM	2025	Preference-aware single ARM	Yes (pref vector)	Yes	Yes
8	PAD	2025	Personalized token reward	Per-user	Yes	Yes
9	Collab	2025	Q-function-based agent switching	Yes	Yes	Yes
10	Darwin	2024	Evolutionary tree search + beam replacement	No	No	Yes
11	RSD	2025	Process-reward rejection (speculative)	No	No	Yes
12	GSI	2025	Soft best-of-N over speculative samples	No	No	Yes
13	Nudging	2025	Small-aligned model token handoff	No	Model-level	Yes
14	SteerLM	2023	Attribute labels in prompt (SFT)	Partial	No	No
15	Arithmetic Ctrl.	2024	Directional pref. w/ reward vectors	Yes	Partial	No
16	ECLIPTICA/CITAN (L1)	2025	Instruction-conditioned policy	Yes	No	No
17	SKlighthouse (L2)	2026	Tournament over K draft candidates	Yes (active blade)	Yes (pointer)	Yes

Table 1: The decode-time alignment landscape. Rows 1–16 are prior work; row 17 is the proposed method. Note that rows 5, 6, 7, 9 already claim modular multi-objective hot-swap.

2.2 The three closest rivals, in detail

The methods Swiss Knife must defeat (or at least differentiate from) are MOD, GenARM, and Collab. Table 2 breaks down the mechanical differences precisely.

Dimension	MOD	GenARM	Collab	Swiss Knife
Core operator	Linear mixture of logits from K full LMs	Weighted sum of ARM rewards added to base logits	Q-function policy switch per token	Discrete tournament over sampled candidates
Per-step compute	All K objective LMs run	Base LM + all K ARMs	All K agent LMs run	One draft + one <i>active</i> auditor
Objective mode	Continuous blend, Pareto front	Continuous blend	Per-token agent selection	Single blade active; α blends with fluency
Degeneracy defense	None explicit	None explicit	Implicit Q-function smoothing	Condorcet aggregation + draft-likelihood tie-break
Speculative framing	No	No	No	Yes (Option B, Sec. 6)

Table 2: Swiss Knife vs. its three closest rivals. Rows in bold type indicate the dimensions where Swiss Knife is genuinely different, not merely rebranded.

Caveat. Reviewers familiar with MOD will ask: “Is Swiss Knife not just MOD with a one-hot weight vector w ?” The answer must cover three mechanical differences: (i) MOD combines full distributions every step, Swiss Knife runs one auditor on sampled candidates; (ii) MOD has no degeneracy defense, Swiss Knife’s tournament structure provably resists single-reward domination; (iii) tournament aggregation is not argmax over linear combination — it uses only *pairwise differences*, which are robust to auditor miscalibration in ways absolute scores are not.

3 What Swiss Knife Actually Contributes

After cross-referencing with the literature, the four-part novelty claim from the original plan document should be restated:

- ✗ “Decode-time alignment with modular heads.” *Taken by MOD, GenARM, PARM, Collab.*
- ✗ “Hot-swap objectives post-deployment.” *Taken by MOD (reweight w), GenARM (reweight ARMs).*
- ✗ “Speculative decoding for efficiency.” *Taken by RSD, GSI — unless actually integrated into verifier (Option B).*
- ✓ **Tournament selection as alternative to linear/weighted combination.** *Genuinely new.*
- ✓ **Tournament structure as built-in defense against reward-max degeneracy.** *Genuinely new, theoretically defensible.*
- ✓ **Single-blade-active switching (not Pareto blending).** *Framing difference that matches real deployment scenarios.*

Three defensible contributions is a solid paper. Four was dilution.

4 Tournament Sampling: The Mechanics

4.1 Why pairwise comparisons?

DPO reward models are well-known to be miscalibrated on absolute scale: the log-ratio $\beta \log[\pi_{\text{blade}}/\pi_{\text{ref}}]$ is a valid preference signal but not a calibrated reward. Argmax-based methods (ARGS, DeAL) use absolute scores and are therefore vulnerable to calibration drift — a blade that assigns inflated scores to "I cannot" tokens will collapse the output distribution to refusals.

Pairwise comparisons ("is A better than B ?") use only *differences* of scores, which are invariant under additive constant shifts:

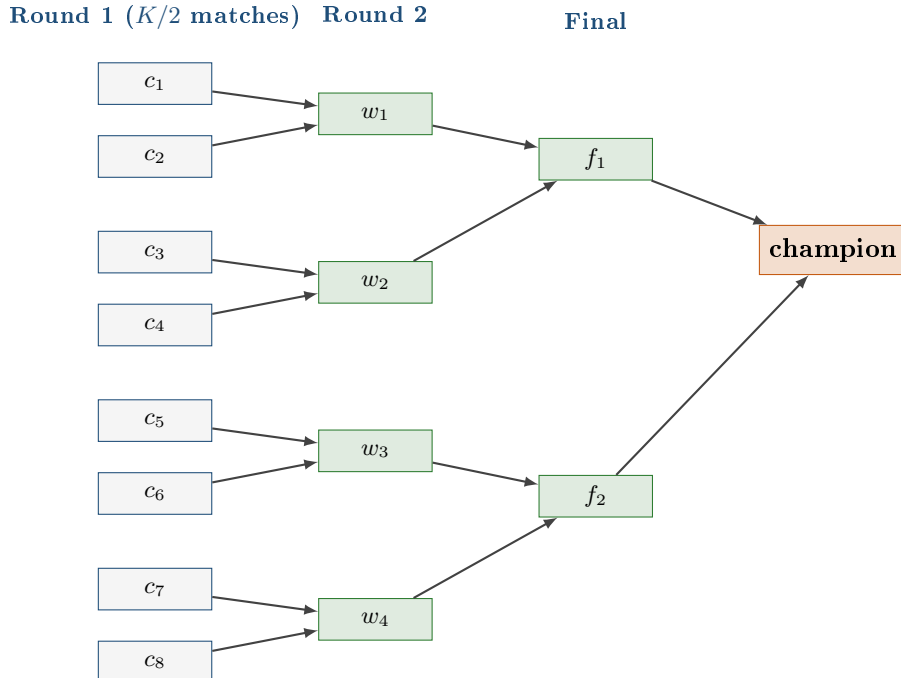
$$\Delta r_{\text{blade}}(A, B) = r_{\text{blade}}(A) - r_{\text{blade}}(B)$$

A blade that is miscalibrated by a constant offset produces $\Delta = 0$ on everything, so the draft/-target likelihood term takes over. This is the mechanical reason tournaments are more robust than argmax.

Key insight. Tournament selection uses only *differences* of scores. This makes it invariant to additive miscalibration of the auditor, which argmax-based methods cannot claim. The paper’s theoretical backbone rests on this observation.

4.2 Format 1: Knockout bracket

A single-elimination bracket. Given K candidates, the tournament runs $\log_2 K$ rounds of parallel pairwise matches. Each round halves the field. Total matches: $K - 1$. Sequential depth: $\log_2 K$.



Pros. Minimal auditor calls, trivially parallel within rounds.

Cons. Pairing is fragile — a strong candidate can be eliminated in round 1 if paired with another strong candidate. Noisy auditors hurt knockout disproportionately.

4.3 Format 2: Swiss-system schedule

This is the format the original plan document mentions without explaining. It is borrowed from competitive chess/go tournaments and solves exactly the weakness of knockout: *no candidate is eliminated after a single loss*. Every candidate plays every round.

4.3.1 How Swiss-system works

Each candidate carries a cumulative score (number of match wins, possibly weighted). The pairing rule is: *in each round, match candidates with the same or closest cumulative score*.

Concrete example. $K = 8$ candidates, $R = 3$ rounds.

#	Candidate	Round 1		Round 2		Round 3		Final score
		Opponent	Result	Opponent	Result	Opponent	Result	
1	c_1	c_2	W (1)	c_3	W (2)	c_5	W (3)	3
2	c_2	c_1	L (0)	c_4	W (1)	c_6	W (2)	2
3	c_3	c_4	W (1)	c_1	L (1)	c_2	L (1)	1
4	c_4	c_3	L (0)	c_2	L (0)	c_8	W (1)	1
5	c_5	c_6	W (1)	c_7	W (2)	c_1	L (2)	2
6	c_6	c_5	L (0)	c_8	W (1)	c_2	L (1)	1
7	c_7	c_8	W (1)	c_5	L (1)	c_3	W (2)	2
8	c_8	c_7	L (0)	c_6	L (0)	c_4	L (0)	0

After 3 rounds, c_1 has 3/3 wins, clearly the strongest. Candidates c_2 , c_5 , c_7 all have 2 wins, broken by tie-break rules.

4.3.2 Pairing rule in each round

1. Group candidates by current cumulative score.
2. Within each score group, pair adjacent candidates (by some secondary order — e.g., initial draft rank).
3. If a score group has odd size, one candidate “floats” to pair with the closest candidate in the next score group. This rarely matters for small K .
4. Avoid repeat pairings when possible (standard Swiss constraint).

4.3.3 Complexity

For K candidates and R rounds: $R \cdot K/2$ matches total. Typical choice $R = \lceil \log_2 K \rceil$:

- $K = 4, R = 2$: 4 matches (vs. knockout: 3).
- $K = 8, R = 3$: 12 matches (vs. knockout: 7).
- $K = 16, R = 4$: 32 matches (vs. knockout: 15).

Roughly $2\times$ the matches of knockout, but considerably more robust to auditor noise because every candidate’s final score reflects multiple matches.

4.3.4 Why use Swiss-system over knockout for Swiss Knife

The name “Swiss Knife” is actually motivated by the Swiss-system pairing: the paper’s branding hinges on this format doing something knockout cannot. That “something” is:

- **Robustness to auditor noise.** A candidate that loses one close match still has chances to recover.

- **Better at selecting “second-best” candidates cleanly.** Useful for diversity metrics (Self-BLEU).
- **Tie-break structure that can incorporate draft likelihood as secondary criterion.**

The experimental claim to defend is: *knockout is a good default; Swiss-system wins when the auditor is miscalibrated or when K is large (≥ 16).* If this is not true in experiments, Swiss-system is dead weight and the paper should drop it.

Caveat. If knockout works just as well as Swiss-system across all tested K and auditor conditions, the Swiss-system format becomes unjustified and the paper’s name becomes awkward. Run this ablation first, not last.

4.4 The match score function

Regardless of tournament format, each individual match between candidates A and B uses the same score:

$$\text{MATCH}(A, B) = \alpha \cdot \left[\log \pi_{\text{draft}}(A) - \log \pi_{\text{draft}}(B) \right] + (1 - \alpha) \cdot \left[r_{\text{blade}}(A) - r_{\text{blade}}(B) \right]$$

with A winning iff $\text{MATCH}(A, B) > 0$.

Here $\alpha \in [0, 1]$ is a mixing coefficient ($\alpha \approx 0.5$ as the default), and r_{blade} is the DPO-implicit reward:

$$r_{\text{blade}}(y|x) = \beta \cdot \log \frac{\pi_{\text{blade}}(y|x)}{\pi_{\text{ref}}(y|x)}$$

This formulation has three virtues:

1. Uses only *differences* — robust to additive auditor miscalibration.
2. Uses your existing DPO blades with zero additional training — no DeBERTa auditor to train.
3. Combines fluency (draft likelihood) and alignment (blade reward) linearly, so α is a well-defined knob.

5 Option A: Non-Speculative (Best-of- K Tournament)

This is the algorithm described implicitly in the original plan document: at each span, the draft samples K candidates, a tournament picks one, commit and loop.

Algorithm 1 Swiss Knife (Option A) — Non-speculative tournament decoding

Require: prompt x , active blade b , mixing α , candidates K , span length L

```

1:  $y \leftarrow []$ 
2: while not done do
3:    $\mathcal{C} \leftarrow \text{draft.sample}(x \oplus y, \text{num} = K, \text{len} = L)$  ▷  $K$  independent spans
4:    $s_d \leftarrow \text{draft.logprob}(\mathcal{C} \mid x \oplus y)$  ▷ batched
5:    $s_b \leftarrow \text{blade.logratio}(\mathcal{C} \mid x \oplus y)$  ▷ batched;  $\log(\pi_b/\pi_{\text{ref}})$ 
6:    $w \leftarrow \text{KNOCKOUTBRACKET}(\mathcal{C}, s_d, s_b, \alpha)$ 
7:    $y \leftarrow y \oplus \mathcal{C}[w]$ 
8: end while
9: return  $y$ 

```

```

1 def knockout_bracket(candidates, draft_scores, blade_scores, alpha):
2     """
3     candidates:      list of K candidate spans (strings or token ids)
4     draft_scores:    tensor of shape [K]          log pi_draft(span | x)
5     blade_scores:    tensor of shape [K]          log pi_blade(span | x)
6                     - log pi_ref(span | x)
7     alpha:           float in [0, 1]             draft-vs-blade mixing
8                     weight
9     returns:         int                         index of winning
10                    candidate
11    """
12    active = list(range(len(candidates)))
13    while len(active) > 1:
14        next_round = []
15        for i in range(0, len(active), 2):
16            a, b = active[i], active[i + 1]
17            delta_draft = draft_scores[a] - draft_scores[b]
18            delta_blade = blade_scores[a] - blade_scores[b]
19            score = alpha * delta_draft + (1 - alpha) * delta_blade
20            next_round.append(a if score > 0 else b)
21        active = next_round
22    return active[0]

```

Listing 1: Option A knockout bracket, plain Python/PyTorch sketch.

Key insight. Option A is what the original plan actually described. It works, it’s simple, and it’s defensible as “tournament-based decode-time alignment.” But it is *not* speculative decoding in any meaningful sense — it’s Best-of- K with a tournament selection rule.

5.1 What Option A is honest about

- No target model is in the loop. Only draft + auditor.
- No acceptance/rejection of speculative sequences. Just candidate scoring.
- No γ -ahead proposal. Just sample K independent spans.
- Speedup relative to vanilla only via the small-draft + small-auditor cost model.

5.2 Why Option A is strictly weaker than Option B

Option A competes directly with Best-of-N, CARDS, Darwin, and the non-speculative members of the decode-time alignment family. The tournament rule is the only novelty. If a reviewer says “run CARDS with a tournament selector,” you have nothing to push back with beyond “we’re the first.”

6 Option B: Speculative-Decoding-Integrated Tournament

This is what the abstract *claims* Swiss Knife is, and what the name “alignment socket” demands. The tournament sits in the verifier slot of a speculative decoding loop.

6.1 Recap: standard speculative decoding

In Leviathan-style speculative decoding, the draft model proposes γ future tokens in one autoregressive sweep. The target model then does *one* forward pass computing π_{target} for all γ positions. At each position i , the verifier accepts token t_i with probability $\min(1, \pi_{\text{target}}(t_i)/\pi_{\text{draft}}(t_i))$. On the first rejection, resample from $(\pi_{\text{target}} - \pi_{\text{draft}})_+$ and discard the tail. This is *lossless* — the output distribution matches π_{target} exactly.

6.2 The reframing: verifier as alignment socket

Swiss Knife’s claim: replace the Leviathan verifier with a tournament over candidate tokens at each speculative position. The tournament uses π_{target} (for calibrated fluency) and r_{blade} (for alignment).

Algorithm 2 Swiss Knife (Option B) — Speculative tournament verifier

Require: prompt x , active blade b , mixing α , lookahead γ , per-position candidates K

```

1:  $y \leftarrow []$ 
2: while not done do
3:    $\mathcal{D} \leftarrow \text{draft.propose}(x \oplus y, \text{lookahead} = \gamma, \text{top}K = K)$  ▷ shape  $[\gamma, K]$ 
4:    $\ell_t \leftarrow \text{target.logprob\_parallel}(\mathcal{D} \mid x \oplus y)$  ▷ ONE forward pass, shape  $[\gamma, K]$ 
5:    $r_b \leftarrow \text{blade.score\_parallel}(\mathcal{D} \mid x \oplus y)$  ▷ shape  $[\gamma, K]$ 
6:   for  $i = 1$  to  $\gamma$  do
7:      $w_i \leftarrow \text{TOURNAMENT}(\mathcal{D}[i, :], \ell_t[i, :], r_b[i, :], \alpha)$ 
8:     if  $\mathcal{D}[i, w_i] \neq \mathcal{D}[i, 0]$  then ▷ Winner differs from draft’s top-1
9:        $y \leftarrow y \oplus \mathcal{D}[1 : i - 1, 0] \oplus \mathcal{D}[i, w_i]$ 
10:      break ▷ discard tail, restart draft
11:    end if
12:  end for
13:  if no rejection occurred then
14:     $y \leftarrow y \oplus \mathcal{D}[:, 0]$  ▷ accept all  $\gamma$  tokens
15:  end if
16: end while
17: return  $y$ 
```

6.3 What makes this actually speculative

1. **One target forward pass per γ positions.** This is where the throughput advantage lives. Without this, you’re paying γ target passes.
2. **Positional acceptance propagation.** First rejection discards the tail — standard speculative decoding mechanics.
3. **Calibrated likelihoods.** Target model provides π_{target} for the draft-agreement term, not a hand-wavy “fluency proxy.”
4. **Alignment socket.** Blade plugs into the verifier; replace it, alignment objective changes. Hot-swap is a pointer operation.

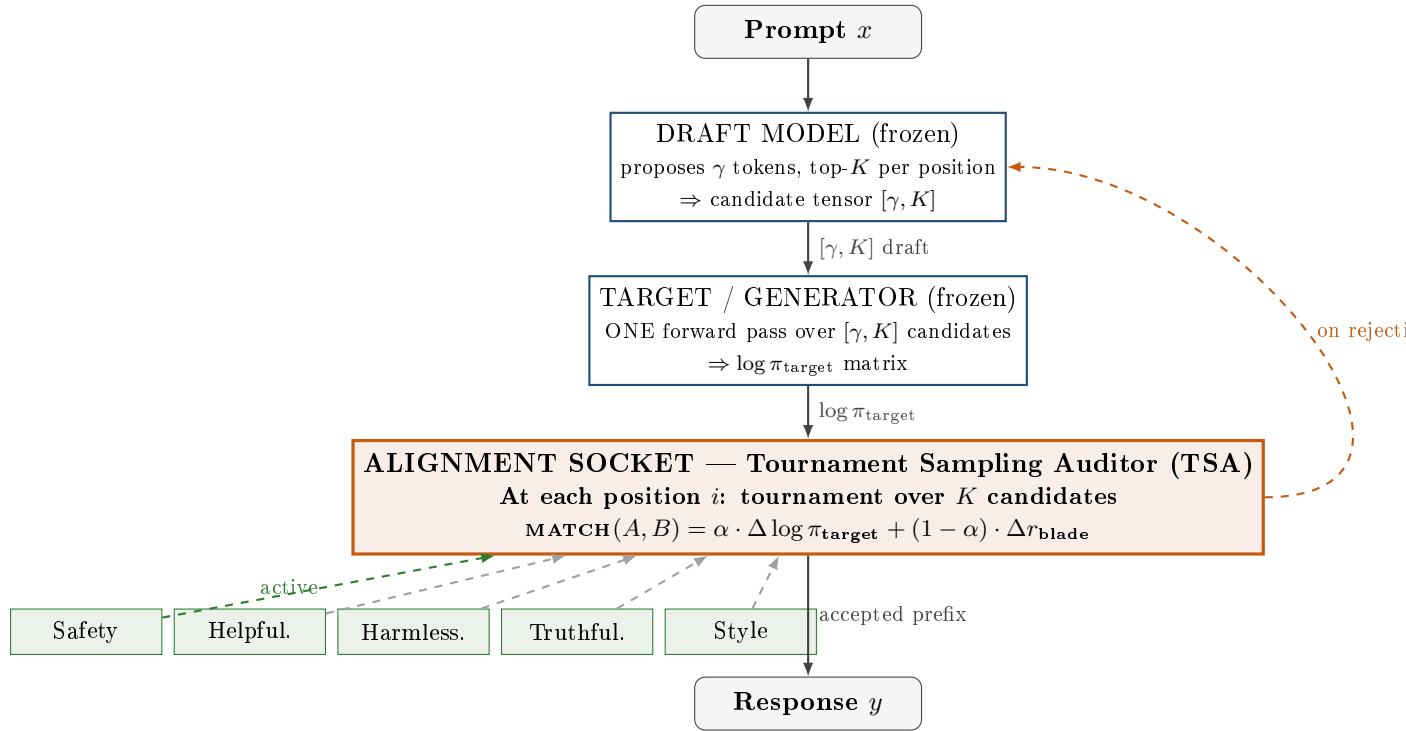
6.4 Compute profile

Per generated span of expected length $\bar{\ell}$:

- Draft forward passes: γ (autoregressive proposal).
- Target forward passes: 1 (parallel verification).
- Blade forward passes: 1 (parallel scoring over $[\gamma, K]$ candidates).
- Tournament matches: $\gamma \cdot (K - 1)$ if knockout, $\gamma \cdot (K/2) \cdot R$ if Swiss.

Throughput relative to vanilla autoregressive target: $\approx 0.5\text{--}0.8\times$ depending on γ and acceptance rate. Matches the envelope reported by RSD.

6.5 Architecture diagram



The socket is where the standard Leviathan verifier would sit. In Swiss Knife, a blade plugs into that slot; different blades produce different alignment behaviors. This is the one-sentence pitch made concrete.

7 Fitting Swiss Knife into the Alignment Framework

The evolutionary narrative the lab has been using:

Level	Approach	Example methods	Where alignment lives
0	Static alignment	DPO, PPO, RLHF, GRPO, ORPO	Baked into weights; one policy per checkpoint
1	Instruction-conditioned switching	ECLIPTICA/CITA, SteerLM	Inside the backbone, conditioned on prompt
2a	Decode-time distribution mixture	MOD, GenARM, PARM	In the logit/reward combiner
SKlight 2b	Decode-time tournament	Swiss Knife	In the verifier slot of a speculative-decoding loop

Key insight. Swiss Knife is *not* a new level above L1. It is a refinement of L2 that already exists. The honest pitch: “MOD and GenARM put alignment in the logit combiner; we put it in the speculative-decoding verifier slot and make the combination rule a tournament instead of a linear mixture.”

The three things Option B gains over Option A and over the L2a methods:

1. **Throughput.** One target forward pass checks γ positions.
2. **Rigor.** Calibrated likelihoods from target, not hand-wavy “fluency proxy.”
3. **Identity.** “We reframed the verifier” is a clean one-sentence pitch that MOD and GenARM cannot claim.

8 Implementation Plan

8.1 The one-sentence novelty claim

Write this on paper, now:

Swiss Knife replaces the Leviathan-style distribution-matching verifier in speculative decoding with a tournament aggregation over per-position candidates, scored by a pluggable alignment blade, yielding a decode-time alignment method that is switchable (single active blade), robust to auditor miscalibration (pairwise comparisons), and resistant to reward-max degeneracies (tournament structure prevents single-candidate domination).

If that sentence survives the experiments, the paper exists. If it doesn't, it doesn't.

8.2 Revised priority order

The original plan document priorities the CITA comparison early. This is wrong. MOD is a stronger rival than CITA for Swiss Knife's specific claim. MOD must be beaten first.

Week	Task	Goal / deliverable
1	Pipeline v0, Option A	Working knockout bracket with 3 DPO blades. Sanity check: outputs are coherent and behave like the active blade.
1–2	MOD vs. TSA bake-off	Same 3 DPO blades, same prompts. Config A: MOD with one-hot w . Config B: Swiss Knife TSA. Measure: cross-obj separation, refuse-always rate, Self-BLEU, latency. This is the paper's life-or-death experiment.
2–3	Degeneracy battery	Refuse-always (500 benign prompts) and Self-BLEU vs. argmax-reward AND vs. MOD linear mix.
3–4	Cross-objective separation matrix	3×3 matrix on ECLIPTICA's 300 prompts. Diagonal $\gg$ off-diagonal is the claim.
4–5	Option B upgrade	Integrate into speculative decoding loop properly. Measure acceptance rate, auditor calls/token, end-to-end latency.
5–6	Tournament format ablation	Knockout vs. Swiss-system, $K \in \{4, 8, 16, 32\}$. If Swiss-system never wins, drop it.
6–7	α sweep and span-length ablation	Find operating point; confirm span-level $>$ token-level for tournament.
7–8	Adversarial battery	HarmBench + GCG attacks on Safety-analog blade.
8–10	ECLIPTICA comparison and paper writing	Final ablation table; draft submission.

8.3 Decision points

- **If TSA beats MOD on degeneracy but matches on alignment:** Lead the paper with *robustness*. Title: "Tournament-Based Decoding for Robust Multi-Objective Alignment."
- **If TSA matches MOD on degeneracy but wins on latency:** Lead with *systems/efficiency*. Positioning: "one-active-blade vs. all-models-every-step."
- **If TSA loses to MOD on both:** Pivot to the deployment story (single-objective guarantees for safety-critical apps) or drop the multi-objective framing entirely and pitch as a robust single-objective decode-time method.
- **If TSA wins on both:** Full Swiss Knife paper; submit to NeurIPS/ICLR.

8.4 The question reviewers will ask, with your answer

Q. Why is tournament selection better than linear combination of token-level rewards?

A. Three reasons:

1. *Theoretical:* Linear combination uses absolute scores and is vulnerable to additive miscalibration. Tournament uses only pairwise differences, which are invariant to constant shifts in auditor output.
2. *Empirical:* Our refuse-always and Self-BLEU results (Weeks 2–3) show tournament resists the degenerate-solution collapse that plagues argmax-reward decoding on the same blades.
3. *Practical:* Single-active-blade switching matches real deployment scenarios. A safety-critical app does not want a 60%-safety-40%-helpfulness Pareto blend; it wants safety mode or helpfulness mode, with the ability to swap without redeployment.

This answer must be defensible on day one of the rebuttal. Write it into the introduction now; the rest of the paper follows.

9 Summary Table: What to Do Next Week

#	Action	Why
1	Stop writing. Implement the 50-line knockout function from §4.	Until the pipeline runs end-to-end on one blade, further design discussion is speculation.
2	Verify DPO log-ratio computation on synthetic A/B pairs.	If the match rule doesn't prefer the clearly-better candidate in a toy test, nothing downstream works.
3	Commit to span-level ($n = 5-10$), not token-level.	Token-level puts you in latency competition with GenARM/RSD which you'll lose.
4	Commit to Option B (speculative verifier integration).	Without it, the abstract's "alignment socket" framing is a marketing lie.
5	Run the MOD bake-off as experiment #1.	It's the paper's existence test. Failing it early saves 6 weeks.
6	Decide knockout vs. Swiss-system empirically, not rhetorically.	If Swiss-system never beats knockout in your setup, the name "Swiss Knife" is aesthetic, and the paper should say so.
7	Kill the small-draft setup (1B draft).	Use ≥ 7 B draft or self-speculative 8B. Small-draft zone is crowded.
8	Write the 3-reason answer to the reviewer Q from §8.4 before submission.	If you can't write it, the paper isn't ready.

The bottom line

Swiss Knife is a real paper if and only if tournament selection provably beats linear combination on degeneracy metrics. Everything else — the blade metaphor, the speculative socket framing, the hot-swap pitch — is supporting scaffolding. Build the bake-off first, decide whether the paper exists, then design the rest.